

Decision Tree

Assignment



Dataset Link : [Bank_Marketing](#)

Submission Guidelines

Students must submit:

- Code (Colab / Jupyter Notebook)
- Outputs (results)
- Proper interpretation for each step

Section 1: Theoretical Questions

Q1. What is a Decision Tree? Explain its structure (Root, Nodes, Leaves) with a real-life example.

Q2. Differentiate between Gini Impurity and Entropy.
Which one is used by default in Scikit-learn and why?

Q3. What is Overfitting in Decision Trees?
How can we detect it using training and testing accuracy?

Q4. Explain Pruning in Decision Trees.
What is the difference between Pre-pruning and Post-pruning?

Q5. What is Feature Importance?
How can it help businesses in decision-making?

Section 2: Practical Questions (Hands-on using your dataset)

Q6. Data Understanding

- Load the dataset
- Display first 5 rows
- Check:
 - Shape
 - Data types
 - Column names

Interpret:

- What types of variables are present?

Q7. Data Cleaning

- Check for missing values
- Handle missing values (if any)
- Check for duplicates

Interpret:

- Is the dataset clean? What issues did you find?

Q8. Data Preprocessing

- Convert categorical variables into numerical
- Mention which encoding method you used

Interpret:

- Why is encoding necessary for Decision Trees?

Q9. Feature Selection & Splitting

- Separate:
 - Features (X)
 - Target (y)
- Perform train-test split (80-20)

Interpret:

- Why is train-test split important?

Q10. Model Building

- Train a Decision Tree Classifier
- Use:
 - `criterion='gini'`
 - `max_depth=5`

Interpret:

- What does `max_depth` control?

Q11. Model Evaluation

Perform:

- Accuracy score
- Confusion Matrix
- Classification Report

Interpret (MUST):

- Is the model performing well?
- Explain Precision & Recall
- Any class imbalance issue?

Q12. Overfitting Check

- Calculate:
 - Training accuracy
 - Testing accuracy

Interpret:

- Is the model overfitting or underfitting?

Q13. Pruning Experiment

- Train another model with:
 - `max_depth=3`
 - `min_samples_split=20`

Interpret:

- Compare performance with previous model
- Which one is better and why?

Q14. Feature Importance

- Extract top 5 important features
- Plot feature importance graph

Interpret:

- Which features influence prediction the most?

Q15. Business Insights

Based on your model:

Answer:

- Which type of customers are more likely to say “yes”?
- What strategy should a bank use



SKILLS